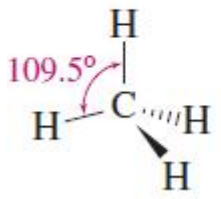
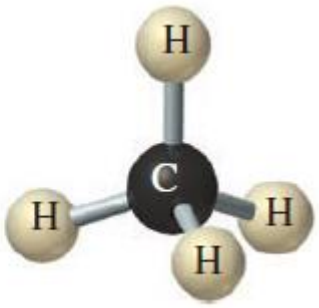
ISOMERISM IN ORGANIC COMPOUNDS

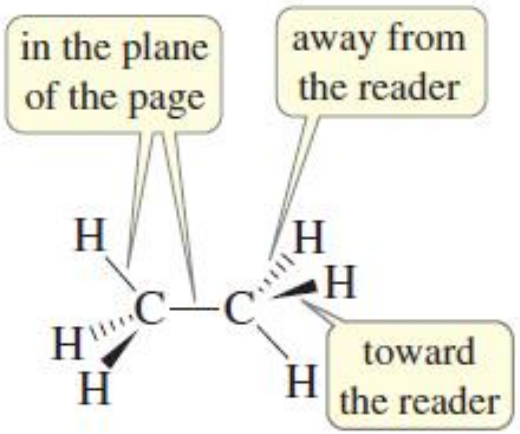
Three-Dimensional Structures of Molecules

- Remember that all molecules are three-dimensional.
- To draw a three-dimensional representation of an organic molecule, chemists use dashed, wedged, and straight lines.

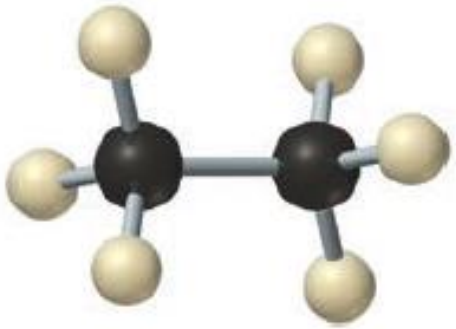




Dashed lines ( )indicate bonds that go backward, away from the reader.
Heavy, wedge-shaped lines ( ) depict bonds that go forward, towards the reader.
Straight lines ( )indicate bonds that are in the plane of the paper.



ethane



ethane

Isomers

- Isomers are different compounds with the same molecular formula.

- There are several types of isomers in organic chemistry.

- The two large classes of isomers are:

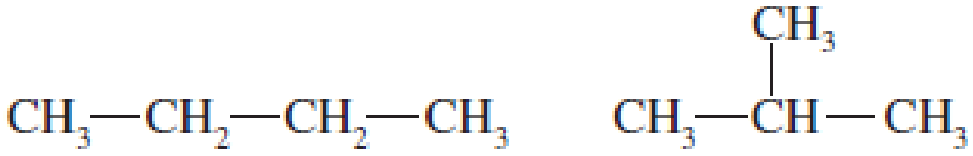
i. Constitutional isomers

ii. Stereoisomers

Constitutional Isomers

- **Constitutional isomers** (or **structural isomers**) are isomers that differ in their bonding sequence, that is, their atoms are connected differently, *but molecular formula is the same*.

- For example, the branching of an alkane.



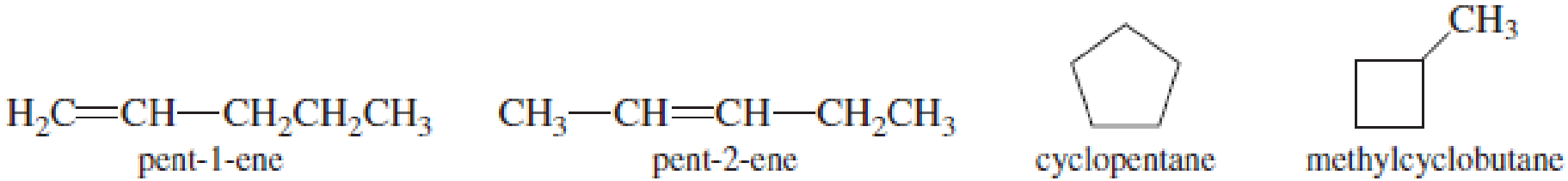
butane 2-methylpropane

- Isomers have different physical and chemical properties.

- Constitutional (structural) isomers may differ in ways other than the branching of the chain.

- They may differ in the position of a double or triple bond, or other functionality, or by having a ring:

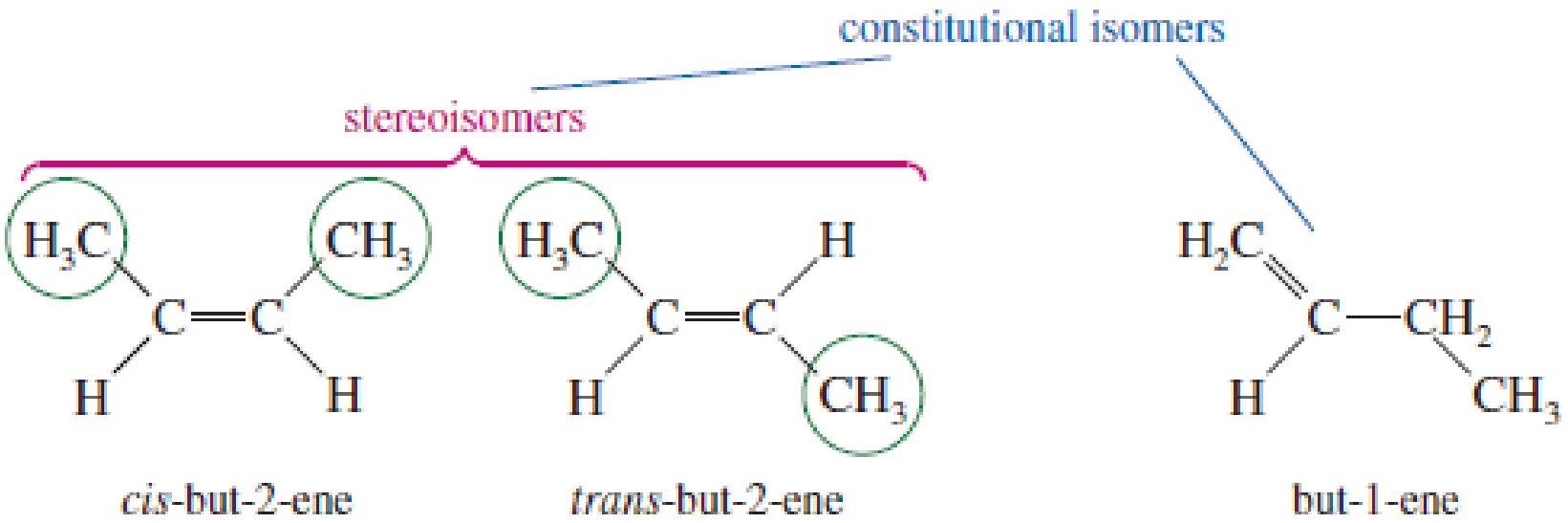
- For example, pent-1-ene, pent-2-ene, cyclopentane, and methylcyclobutane are all structural isomers of each other:



Molecular formula: C_5H_{10}

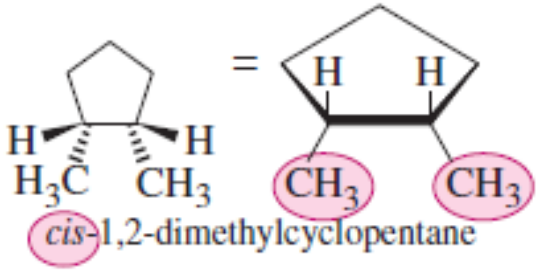
Stereoisomers

- Stereoisomers are isomers that differ only in how their atoms are arranged in space.
- I.e. same connection of atoms and bonds, but different spatial arrangement of the atoms.
- An important class of stereoisomers are the ***geometric isomers***.
- For example *cis*- and *trans*-2-butene are geometric isomers, because they have the same connectivity of bonds, but different spatial orientation of the groups attached to the double bond.

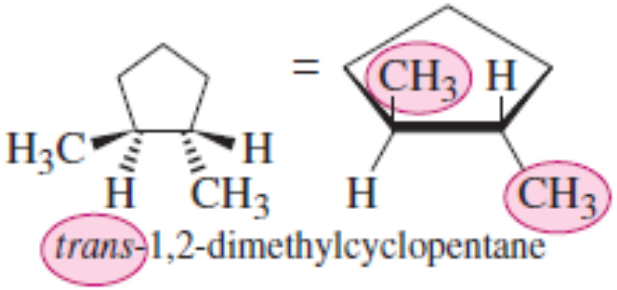


But-1-ene and but-2-ene are constitutional isomers.
Cis-but-2-ene and *trans*-but-2-ene are stereoisomers (geometric isomers).

- Due to rigidity of their bonds, cycloalkanes also exhibit *cis*- and *trans*-isomerism.
- A cycloalkane has two distinct faces, the upper face and the lower face.
- If two same substituents point towards the same face, it is known as a *cis* isomer.

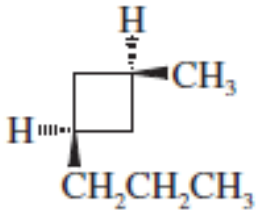


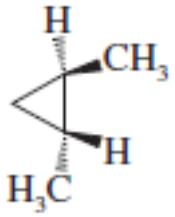
- If two same substituents point towards opposite faces, it is known as a *trans* isomer.



Practice exercise 1:

Give IUPAC names for the following cycloalkanes:



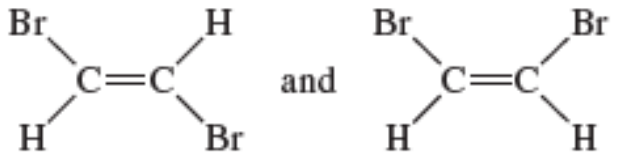


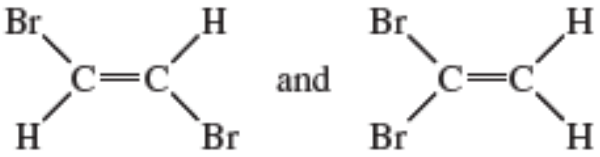
E/Z isomers are also geometric isomers (see Lecture 2)

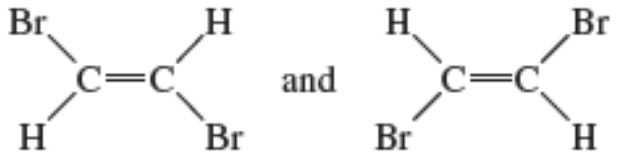
Practice exercise 2:

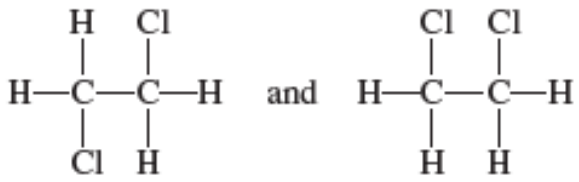
Give the relationship between the following pairs of structures. The possible relationships are: *same compound; constitutional (structural) isomers; geometric isomers; not isomers.*

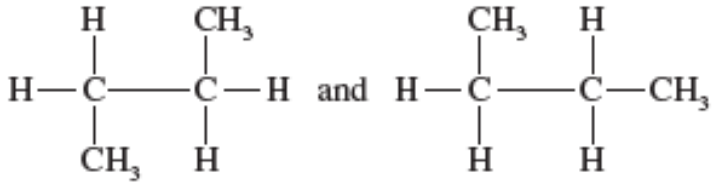
(a) $CH_3CH_2CH(CH_2CH_3)CH_2CH_3$ and $CH_3CH_2CH(CH_3)CH_2CH_2CH_3$

(b) 

(c) 

(d) 

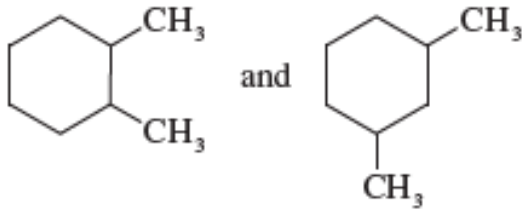
(e) 

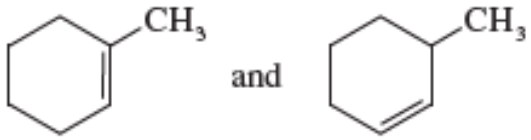
(f) 

(g) $CH_3—CH_2—CH_2—CH_3$ and $CH_3—CH=CH—CH_3$

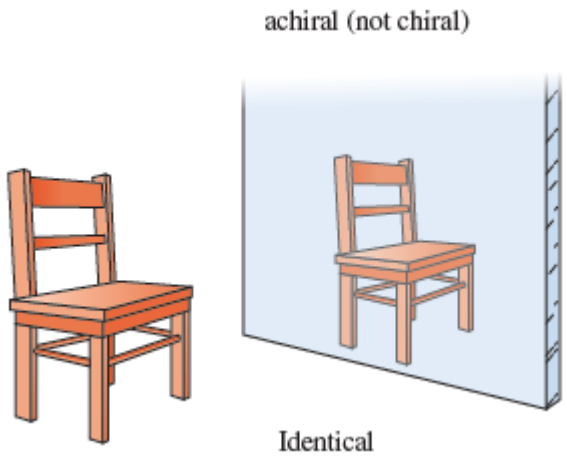
(h) $CH_2=CH—CH_2CH_2CH_3$ and $CH_3—CH=CH—CH_2CH_3$

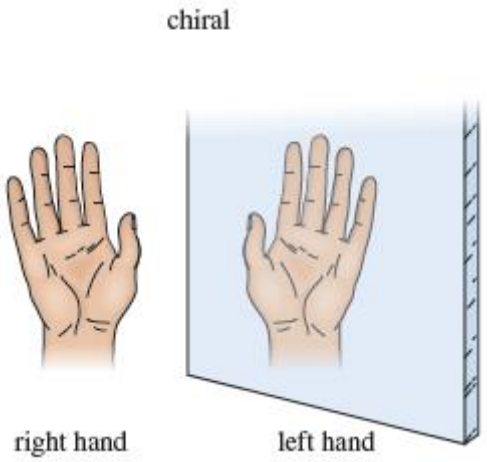
(i) $CH_2=CHCH_2CH_2CH_3$ and $CH_3CH_2CH_2CH=CH_2$

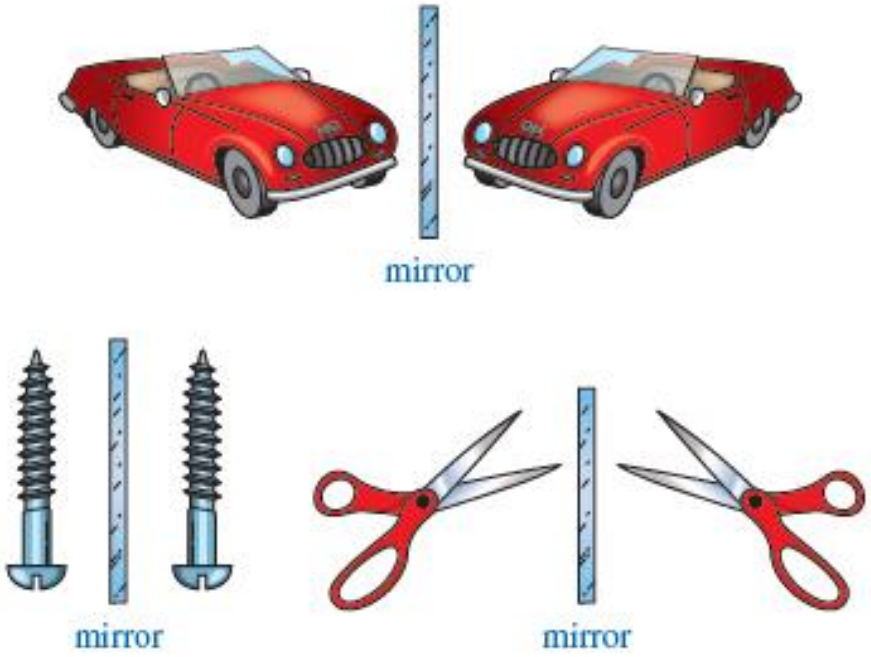
(j) 

(k) 

- Another group of stereoisomers are the ***chiral isomers***.
- Objects that are non-superimposable mirror images of each other are said to be *chiral*.
- For example, your left hand and your right hand are non-superimposable mirror images of each other – therefore they are chiral.
- Objects that have mirror images that are superimposable are said to be *achiral*.







Chiral objects

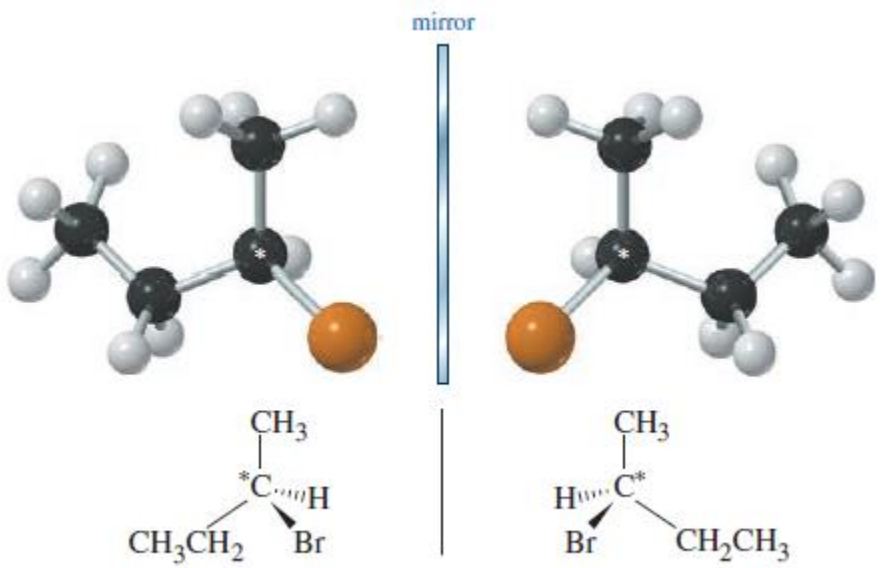
- Like other objects, molecules are either chiral or achiral.

- Two molecules are said to be **superimposable** if they can be placed on top of each other and the three dimensional position of each atom of one molecule coincides with the equivalent atom of the other molecule.

- To draw the mirror image of a molecule, simply draw the same structure with left and right side of the molecule reversed.

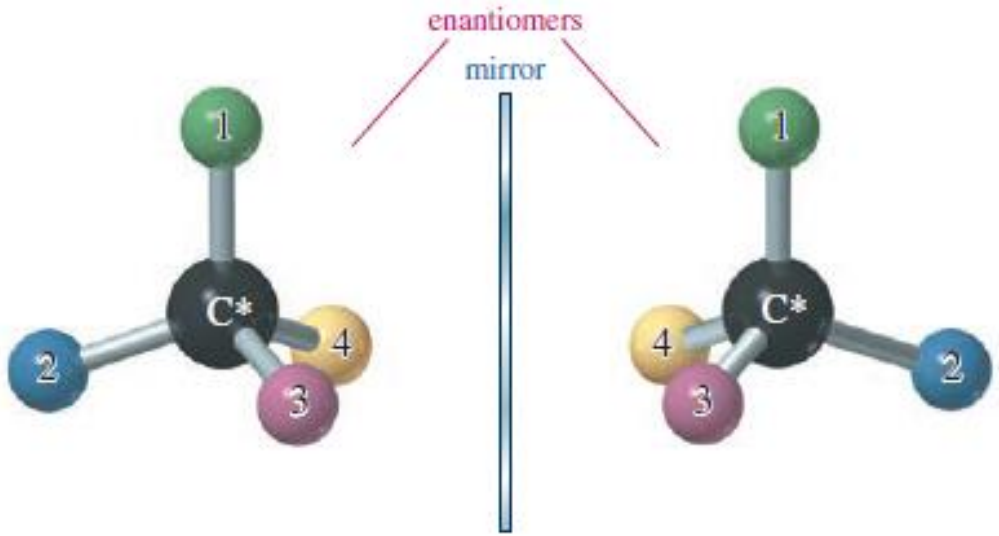
- The mirror image isomers of a chiral molecule are known as *enantiomers*.

- How do you know if a molecule is chiral or not?
- The most common feature that leads to chirality is a **carbon atom that is bonded to four different groups**.
- Such a carbon atom is known as an asymmetric carbon atom, or a chiral carbon atom, and is designated with an asterix (*)



The 2 enantiomers of 2-bromobutane

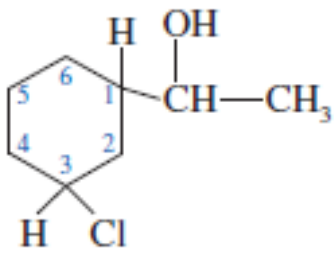
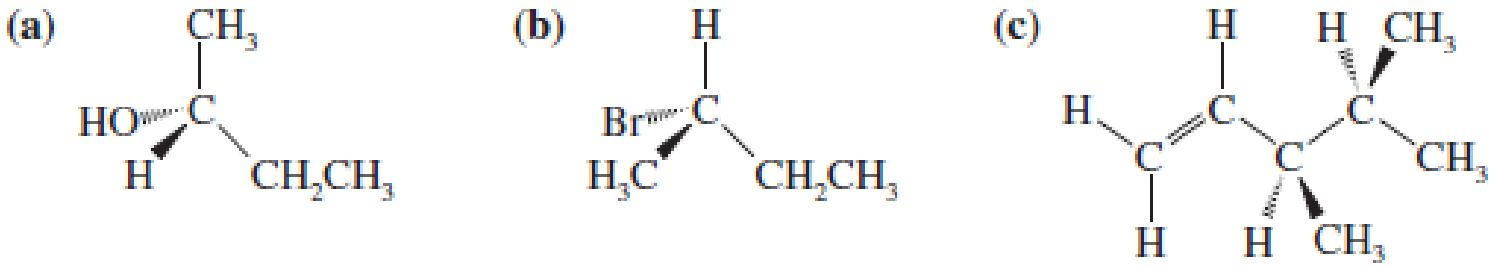
Asymmetric carbon, therefore molecule must be chiral.



Non-superimposable mirror images of each other – chiral

- Practice exercise 3:

Identify each asymmetric carbon in the following structure:



Identify the asymmetric carbons in each of the following structures, using the asterix symbol, and draw its mirror image (enantiomer).